Mini Project Exhibition

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NexusBus – Real-Time Bus Tracking System

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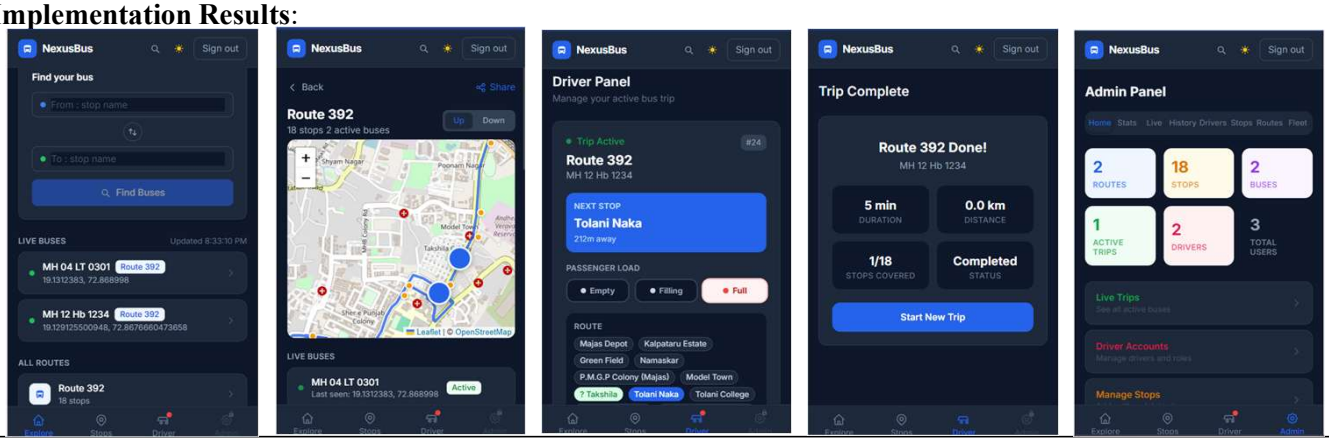
Abstract: NexusBus is a real-time bus tracking web application designed to improve public transport visibility. The system allows passengers to track buses live on a map without logging in, while drivers automatically send GPS location updates during trips. Admins can manage routes, buses, stops, and drivers through a centralized dashboard. The application uses modern web technologies and interactive maps to provide accurate bus locations and improve commuter experience.

Keywords: Real-Time Tracking, GPS, React, Spring Boot, OpenStreetMap, Smart Transport.

Problem Statement: Passengers waiting at bus stops often do not know when the next bus will arrive or where the bus is currently located. This lack of real-time information leads to long waiting times and frustration. NexusBus addresses this problem by providing a real-time bus tracking system that displays live bus locations and route information.

Technology Stack: Frontend: React, Vite, Tailwind CSS ; Backend: Java, Spring Boot ; Database: PostgreSQL ; Maps: Leaflet, OpenStreetMap ; Deployment: Netlify, Render, Supabase.
Applications: Public transport tracking systems, Smart city transportation management, Bus fleet monitoring for organizations, College and school bus tracking, Urban mobility solutions

Implementation Results:



Conclusion: NexusBus successfully demonstrates a real-time bus tracking system using modern web technologies. The system improves passenger convenience by providing live bus location information and enables administrators to monitor transport operations efficiently.

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